

***In Vivo* Epigenetic CRISPR Screen Identifies *Asf1a* as an Immunotherapeutic Target in *Kras*-Mutant Lung Adenocarcinoma**



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ABSTRACT

Despite substantial progress in lung cancer immunotherapy, the overall response rate in patients with *KRAS*-mutant lung adenocarcinoma (LUAD) remains low. Combining standard immunotherapy with adjuvant approaches that enhance adaptive immune responses—such as epigenetic modulation of antitumor immunity—is therefore an attractive strategy. To identify epigenetic regulators of tumor immunity, we constructed an epigenetic-focused single guide RNA library and performed an *in vivo* CRISPR screen in a *Kras*^{G12D}/*Trp53*^{-/-} LUAD model. Our data showed that loss of the histone chaperone *Asf1a* in tumor cells sensitizes tumors to anti-PD-1 treatment. Mechanistic studies revealed that tumor cell-intrinsic *Asf1a* deficiency induced immunogenic macrophage differentiation in the tumor microenvironment by upregulating GM-CSF expression and potentiated T-cell activation in combination with anti-PD-1. Our results provide a rationale for a novel combination therapy consisting of ASF1A inhibition and anti-PD-1 immunotherapy.

SIGNIFICANCE: Using an *in vivo* epigenetic CRISPR screen, we identified *Asf1a* as a critical regulator of LUAD sensitivity to anti-PD-1 therapy. *Asf1a* deficiency synergized with anti-PD-1 immunotherapy by promoting M1-like macrophage polarization and T-cell activation. Thus, we provide a new immunotherapeutic strategy for this subtype of patients with LUAD.

See related commentary by Menzel and Black, p. 179.

INTRODUCTION

Lung adenocarcinoma (LUAD) is a leading cause of cancer-related mortality worldwide (1). *KRAS* (32%) and *EGFR* (11%) mutations are major oncogenic drivers in LUAD (2). Molecular targeted therapy is a promising therapeutic modality for patients with LUAD compared with conventional chemotherapy or radiotherapy. Patients with LUAD with *EGFR* mutations can benefit from *EGFR* tyrosine kinase inhibitors (3–5). Despite the development of allele-specific

KRAS^{G12C} inhibitors (6–8), *KRAS* remains an elusive target for direct inhibitors (9), highlighting an urgent need to develop new therapeutic strategies for *KRAS*-mutant patients. In this regard, immunotherapy has provided additional treatment options for patients with cancer. Blockade of inhibitory immune-checkpoint receptors such as PD-1 and CTLA4 have achieved clinical success in multiple cancers, including LUAD (10). However, the immunotherapeutic response rate in *KRAS*-mutant patients remains unsatisfactory (11). Combining immunotherapy with adjuvant approaches that enhance adaptive immune responses is therefore a potential strategy (12).

Epigenetic genes play important roles in cancer biology (13), and accumulating evidence indicates that epigenetic factors are involved in modulating the tumor immune microenvironment (TME) and regulating the antitumor immune response. For example, DNA methyltransferases (DNMT), histone deacetylases (HDAC), enhancer of zeste homolog 2 (EZH2), bromodomain-containing 4 (BRD4), and lysine-specific histone demethylase 1A (LSD1) play important roles in cancer biology and can modulate antitumor immunity (14–17). However, the utility of epigenetic regulators in potentiating cancer immunotherapy remains underexplored. CRISPR/Cas9 has expanded the use of functional genetic screens for rapid target discovery in immunotherapy (18–20). Specifically, coculture systems have been used to screen for factors regulating interactions between cancer cells and immune cells (19, 20). However, these *in vitro* screen designs do not faithfully capture the complex interactions that occur within the endogenous tumor microenvironment. *In vivo* models can be a more relevant setting to screen for tumor-immune interactions, but are challenging considering the technical difficulty in maintaining single guide RNA (sgRNA) representation *in vivo* (21). Therefore, using small and focused libraries is often a more practical strategy for *in vivo* CRISPR screens (18).

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Using an epigenetic-focused *in vivo* CRISPR screen in the *Kras*^{G12D}/*Trp53*^{-/-} (KP) LUAD model, we studied the functions of epigenetic genes in modulating antitumor immunity and identified *Asf1a* as a potential therapeutic target. ASF1 is a histone H3–H4 chaperone conserved from yeast to human cells. ASF1A and ASF1B are mammalian isoforms involved in DNA replication-coupled and DNA replication-independent nucleosome assembly pathways (22). ASF1 also plays a role in regulating gene transcription. For example, ASF1A resolves bivalent chromatin domains for the induction of lineage-specific genes during embryonic stem cell differentiation (23). Functional and mechanistic studies showed that *Asf1a* deficiency sensitizes LUAD tumors to anti-PD-1 therapy by promoting M1-like macrophage polarization and enhancing T-cell activation. Our findings provide a rationale for combining ASF1A inhibition and anti-PD-1 immunotherapy in patients with LUAD.

RESULTS

In Vivo CRISPR Screen Identifies Epigenetic Regulators of Tumor Immunity

To systemically assess cell-intrinsic epigenetic regulators of tumor immunity, we developed an *in vivo* CRISPR screen using the KP mutant lung cancer mouse model (Fig. 1A). First, we generated an epigenetic-focused sgRNA library, which included sgRNAs targeting 524 epigenetic regulators and 173 control genes (essential genes, immune modulators), and nontargeting guides (Supplementary Table S1). We confirmed an even distribution of guides (Supplementary Fig. S1A). Next, we generated clonal KP mouse lung cancer cell lines with or without stable expression of Cas9 (Supplementary Fig. S1B), and confirmed Cas9 activity in KP-Cas9 clones (Supplementary Fig. S2A–S2C). We assessed the tumor formation capacity of library-transduced KP-Cas9 clones (Supplementary Fig. S2D and S2E) and evaluated the sgRNA representation in tumors derived from KP clones (no Cas9) using the sgRNAs as barcodes (Supplementary Fig. S2F). KP-Cas9-clone 7 was selected for *in vivo* CRISPR screens because the clone showed superior Cas9 activity (Supplementary Fig. S2A) and maintained the optimal sgRNA representation *in vivo* (Supplementary Fig. S2F). Next, we injected early-passage KP-Cas9-clone 7 library cells into *Rag1*^{-/-} and wild-type (WT) mice. On day 7, mice were treated with anti-PD-1 or isotype control. To identify changes in sgRNA abundances across treatment groups, tumors were harvested on day 24 (12 tumors from 6 mice in each group), genomic DNA was isolated, and amplified sgRNA samples were prepared for next-generation sequencing (NGS). There were 4 treatment groups: immunodeficient *Rag1*^{-/-} mice treated with control IgG (ID-IgG) or anti-PD-1 (ID-PD1), and immunocompetent WT mice treated with control IgG (IC-IgG) or anti-PD-1 (IC-PD1). Comparing sgRNAs recovered from tumors in B6 *Rag1*^{-/-} and B6 WT mice treated with control IgG (ID-IgG vs. IC-IgG), we identified epigenetic targets that, upon loss, modulated the anti-tumor immune response in the immunocompetent WT mice (Supplementary Fig. S3A and S3B; Supplementary Table S2). sgRNAs targeting phosphatase and tensin homolog (*Pten*) were significantly enriched, and sgRNAs targeting bromodomain-containing protein 4 (*Brd4*) were significantly

depleted in anti-PD-1-treated WT hosts (Supplementary Fig. S3A). This is consistent with their dichotomous roles in regulating tumor immunity (24, 25). Comparing sgRNAs recovered from tumors in B6 WT mice treated with control IgG and anti-PD-1 (IC-IgG vs. IC-PD1), we identified targets that, when lost, either enhanced or inhibited sensitivity to anti-PD-1 treatment (Fig. 1B and C; Supplementary Table S2). As expected, sgRNAs targeting genes required for high expression of major histocompatibility complex class I (MHC I; e.g., *Tap1*, *Tap2*, *B2m*, *Stat1*, *Jak1*, or *Jak2*) promoted resistance to immunotherapy and were enriched in tumors treated with anti-PD-1 (Fig. 1B). By contrast, sgRNAs targeting *Ctmb1* and *Mapk3* were significantly depleted (Fig. 1B), consistent with findings that β -catenin or MAPK3 inhibition promotes sensitivity to immune checkpoint blockade (26, 27). Of note, sgRNAs targeting the histone chaperone gene *Asf1a* were also significantly depleted in tumors treated with anti-PD-1 (Fig. 1B–E). Importantly, *Asf1a* sgRNAs were only depleted by anti-PD-1 treatment in WT but not *Rag1*^{-/-} mice and were not depleted by control IgG, suggesting specificity of responsiveness to PD-1-based immunotherapy and an enhanced T-cell response (Fig. 1D). Based on these results, we hypothesized that *Asf1a* promotes suppression of tumor immunity.

Asf1a Deficiency Enhances Sensitivity to Anti-PD-1 Treatment

ASF1A is overexpressed in a variety of primary human tumors, including LUAD, and higher ASF1A expression is associated with a significantly poorer outcome in patients with hepatocellular carcinoma (28). To evaluate the function of *Asf1a* in LUAD, we established KP-Cas9 clones with *Asf1a* knockout (KO; Supplementary Fig. S4A). *Asf1a* KO showed no obvious effect on the expression of its paralog gene *Asf1b* (Supplementary Fig. S4B). ASF1A inhibition can trigger DNA damage (28); however, we found that γ H2AX levels showed no significant difference in KP cells with or without *Asf1a* KO (Supplementary Fig. S4C). To determine whether *Asf1a* KO alters the growth of KP lung tumors, we performed *in vitro* colony formation assays and an orthotopic xenograft experiment. *In vitro* colony formation assays showed no significant effect of *Asf1a* KO on tumor cell proliferation (Supplementary Fig. S4D and S4E). Orthotopic xenograft experiments also demonstrated that *Asf1a* KO had no effect on tumor growth in *Rag1*^{-/-} mice (Supplementary Fig. S4F and S4G). Consistently, using The Cancer Genome Atlas (TCGA) human lung cancer database, we found ASF1A expression shows no significant correlation with patient survival in either LUAD or lung squamous cancer (Supplementary Fig. S4H and S4I). However, using allograft experiments in WT mice we confirmed that *Asf1a* deficiency sensitizes tumors to anti-PD-1 treatment in orthotopic lung cancer models (Fig. 2A–D). Using an inducible shRNA system, we further confirmed *Asf1a* knockdown (KD) exerted only a marginal effect on tumor cell proliferation (Supplementary Fig. S5A–S5C); however, *Asf1a* KD in combination with anti-PD-1 treatment significantly inhibited tumor growth in WT mice (Supplementary Fig. S5D–S5F). We also performed the treatment study in another KP model (KP-2; Supplementary Fig. S6A–S6C). This model

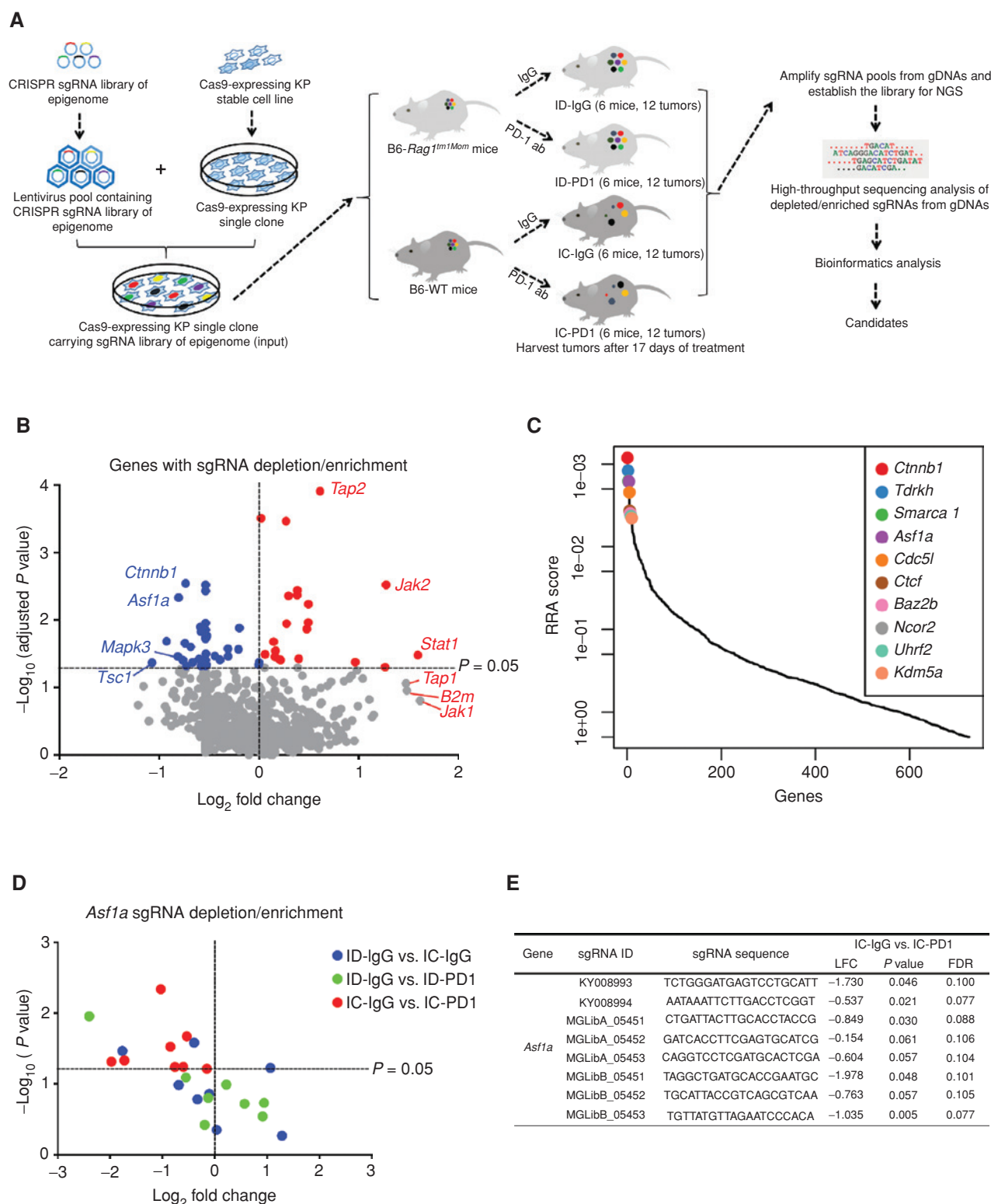


Figure 1. *In vivo* epigenome-wide CRISPR screen identifies *Asf1a* as a negative regulator of response to anti-PD-1 therapy. **A**, Strategy of *in vivo* epigenome-wide CRISPR screen. Twelve tumors from 6 mice were included in each group of the screen. **B**, Volcano plot illustrating the comparison of IC-IgG and IC-PD1 genes whose knockout (KO) can enhance (blue) or inhibit (red) sensitivity to anti-PD-1 treatment. Some top candidates are highlighted, along with positive control genes whose KO is expected to enhance or inhibit anti-PD-1 treatment. **C**, Illustration of the top 10 candidates from **B**. RRA, Robust Rank Aggregation. **D**, Scatter plot showing the performance of 8 *Asf1a* sgRNAs in the comparisons indicated as "ID-IgG vs. IC-IgG," "ID-IgG vs. ID-PD1," and "IC-IgG vs. IC-PD1." **E**, Detailed information on the performance of 8 *Asf1a* sgRNAs in the comparison "IC-IgG vs. IC-PD1." ID, immunodeficient B6 *Rag1*^{-/-} mice; IC, immunocompetent B6 mice; IgG, IgG treatment; PD1, anti-PD-1 treatment.

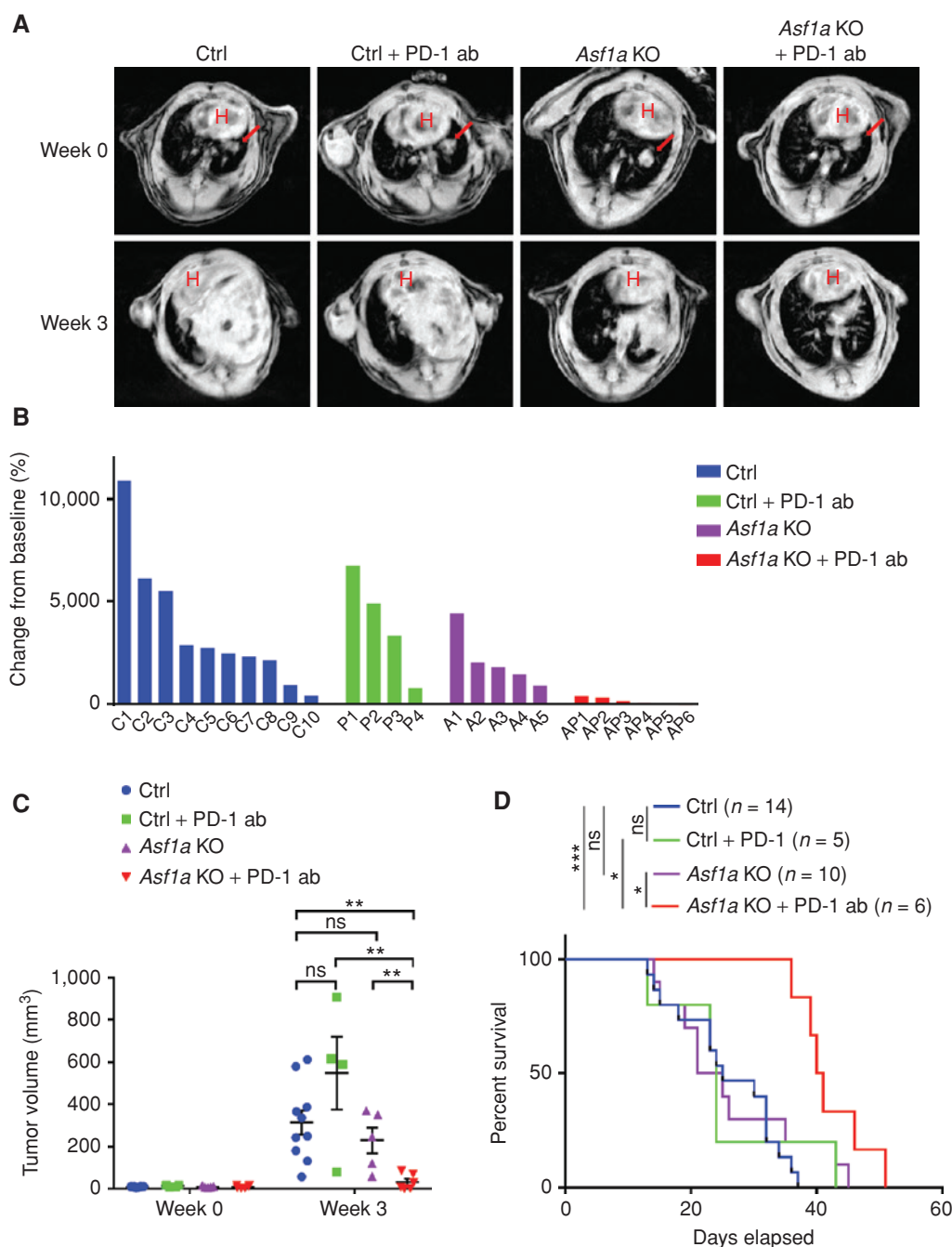


Figure 2. *Asf1a* deficiency synergizes with anti-PD-1 treatment to inhibit tumor progression. **A**, Representative MRI scans (1 of 24 scanned images of each mouse) showing mouse lung tumors before and after treatment. The red arrow indicates the single tumor nodule on the left lobe. "H" indicates the heart. **B**, Waterfall plot showing percentage changes in tumor volume in response to treatment. Each column represents one mouse. **C**, Dot plot illustrating the tumor volume across the different treatment groups (Ctrl, *n* = 10; Ctrl + PD-1 ab, *n* = 4; *Asf1a* KO, *n* = 5; *Asf1a* KO + PD-1 ab, *n* = 6). Four mice in the Ctrl group, 1 mouse in the Ctrl + PD-1 ab group, and 5 mice in the *Asf1a* KO group died prior to week 3 MRI imaging, and hence are excluded here. **D**, Survival curve for each group in the treatment study (Ctrl, *n* = 14; Ctrl + PD-1 ab, *n* = 5; *Asf1a* KO, *n* = 10; *Asf1a* KO + PD-1 ab, *n* = 6). All data are mean ± SEM. *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001.

is very sensitive to anti-PD-1 treatment for as-yet-unclear reasons; however, *Asf1a* KO also showed a significant synergistic effect with anti-PD-1 to inhibit KP-2 allograft tumors (Supplementary Fig. S6C). Collectively, these data confirmed our *in vivo* CRISPR screen results that *Asf1a* deficiency potentiated the effect of anti-PD-1 immunotherapy.

Additionally, we performed the treatment study in a colon cancer model (MC38 allograft model) and found *Asf1a* KO significantly synergized with anti-PD-1 to inhibit MC38 allograft tumors (Supplementary Fig. S6D–S6G), suggesting that ASF1A loss might be a more widespread biomarker for susceptibility to anti-PD-1.

Tumor Cell-Intrinsic *Asf1a* Deficiency Promotes M1-Like Macrophage Polarization and T-cell Activation

To determine the mechanism by which loss of *Asf1a* enhances sensitivity to anti-PD-1, we evaluated the immune profile of orthotopic lung tumors following treatment (Supplementary Fig. S7). Targeting ASF1A or PD-1, alone or in combination, did not induce changes in the prevalence of CD45⁺, CD3⁺, CD4⁺, or CD8⁺ populations in the LUAD tumor microenvironment (Supplementary Fig. S8A–S8D). Similarly, there was minimal correlation between *ASF1A* and *CD3D*, *CD4*, or *CD8A* expression in LUAD samples in TCGA (Supplementary Fig. S8E–S8G). In addition, *Asf1a* KO or anti-PD-1 treatment alone showed a weak effect on the activation of effector T cells (Fig. 3A–D; Supplementary Fig. S8H–S8Q); however, *Asf1a* KO plus anti-PD-1 treatment markedly enhanced adaptive immunity (Fig. 3A–D; Supplementary Fig. S8H–S8Q). Moreover, rechallenge experiments in the MC38 allograft model showed that rechallenged tumor cells grew more slowly in the *Asf1a* KO tumor-bearing mice pretreated with anti-PD-1 compared with control (Ctrl) tumor-bearing mice pretreated with anti-PD-1 (Supplementary Fig. S9A and S9B), which supports the existence of memory T cells in response to *Asf1a* loss. In addition, an *ex vivo* experiment showed that isolated pan T cells from the lungs of the KP-*Asf1a* KO plus anti-PD-1 treatment group showed a stronger cytotoxicity to tumor cells when compared with those from spleens or lungs of Ctrl mice (Supplementary Fig. S9C), suggesting that T cells from the combination group can specifically recognize tumor-associated antigens and result in cancer cell killing.

Furthermore, inflammatory monocyte and macrophage populations were significantly enriched in *Asf1a* KO tumors (Fig. 3E and F), whereas tumor-associated neutrophils (TAN), myeloid-derived suppressing cells (MDSC), eosinophils, CD103⁺ dendritic cells (DC), and alveolar macrophage populations were not obviously affected (Supplementary Fig. S10A–S10G). Analysis of macrophage phenotypes demonstrated that the M1-like macrophages were significantly enriched in *Asf1a* KO tumors (Fig. 3G and H; Supplementary Fig. S10H and S10I). Furthermore, *Asf1a* KO in MC38 cells also significantly promoted transcription of the gene encoding GM-CSF and increased the expression of M1 macrophage markers (IA/IE, CD80, and CD86) in macrophages (Supplementary Fig. S6H–S6J). Collectively, these data suggested that *Asf1a* deficiency sensitizes tumors to anti-PD-1 treatment by promoting M1-like macrophage polarization and T-cell activation. This observation is consistent with our previous observations that M1-like macrophages promote T-cell activation in cancer and enhance sensitivity to PD-1–based immunotherapy (29, 30).

Tumor Cell-Intrinsic *Asf1a* Deficiency Upregulates GM-CSF Expression

To explore the mechanism by which *Asf1a* deficiency promotes M1-like macrophage polarization in LUAD, we performed RNA sequencing (RNA-seq) on *Asf1a* KO and control KP cells. Gene set enrichment analysis (GSEA) revealed that the genes related to the TNFA signaling via NFκB and inflammatory response pathways were significantly enriched in KP

cells with *Asf1a* KO (Fig. 4A–C; Supplementary Table S3). In parallel, we compared *ASF1A*-low LUAD samples with *ASF1A*-high LUAD samples in TCGA, and found TNFA signaling via NFκB and inflammatory response pathways were also significantly enriched in the *ASF1A*-low LUAD samples (Fig. 4D and E; Supplementary Table S3). Of note, GM-CSF (encoded by gene *Csf2*), an inflammatory cytokine in the TNFA pathway that promotes inflammatory monocyte infiltration and M1-like macrophage polarization (31, 32), was significantly upregulated in *Asf1a* KO KP cells (Fig. 4F–H). Similarly, we observed higher levels of GM-CSF in cell culture supernatant from *Asf1a* KO or KD KP cells (Fig. 4I; Supplementary Fig. S11A). Furthermore, analysis of mouse-derived organotypic tumor spheroids (MDOTS) revealed higher GM-CSF levels in *Asf1a* KO MDOTS (Supplementary Fig. S11B). Additionally, *Asf1a* KO in the KP-2 and MC38 cell lines also promotes GM-CSF expression (Supplementary Fig. S6K and S6L).

ASF1A cooperates with CAF1 to promote replication-dependent chromatin assembly and with HIRA to promote replication-independent chromatin assembly (22), and nucleosome dynamics regulate gene transcription (33). CAF1 subunits (*Chaf1a*, *Chaf1b*) and HIRA were included in our sgRNA library, but the knockout of *Chaf1a*, *Chaf1b*, or *Hira* did not sensitize to anti-PD-1 therapy in the screen (Supplementary Table S2). Assay for transposase-accessible chromatin using sequencing (ATAC-seq) data showed DNA accessibility of the *Csf2* gene did not differ among KP cells with or without *Asf1a* knockout (Supplementary Fig. S12A). However, chromatin immunoprecipitation sequencing (ChIP-seq) data in HeLa cells (34) showed that ASF1A occupies the promoter of *CSF2*, the gene encoding GM-CSF (Fig. 4J). Accordingly, *ASF1A* KD in the human lung cancer cell line NCI-H2009 increased GM-CSF expression (Supplementary Fig. S12B and S12C), and ChIP-seq data in this cell line also showed the occupancy of ASF1A on the promoter of *CSF2* (Fig. 4K). These data support that ASF1A may directly inhibit the transcription of GM-CSF, independent of nucleosome assembly.

Asf1a Deficiency Promotes Innate and Adaptive Immunity via GM-CSF

To further evaluate how *Asf1a* deficiency in tumor cells promotes M1-like macrophage polarization, we cocultured tumor cells with bone marrow macrophage precursors. *Asf1a* KO LUAD cells significantly increased M1-like macrophage differentiation (Fig. 5A and B). To determine whether this effect is mediated by GM-CSF, we used a GM-CSF neutralizing antibody in the coculture system, and observed a significantly reduced M1-like macrophage population under these conditions (Fig. 5C and D). These data suggest that tumor-intrinsic *Asf1a* loss promotes M1-like macrophage polarization through upregulation of GM-CSF. To determine whether immunogenic macrophage differentiation, in turn, accounts for the enhanced T-cell activation associated with *Asf1a* loss, we sorted macrophages from the tumor cell–bone marrow macrophage coculture system and performed secondary coculture with T cells. We observed that the macrophages entrained by *Asf1a* KO tumor cells promoted enhanced activation of T cells (Fig. 5E and F; Supplementary Fig. S13A–S13F). Collectively, these data indicate that tumor cell-intrinsic *Asf1a* deficiency enhances M1-like

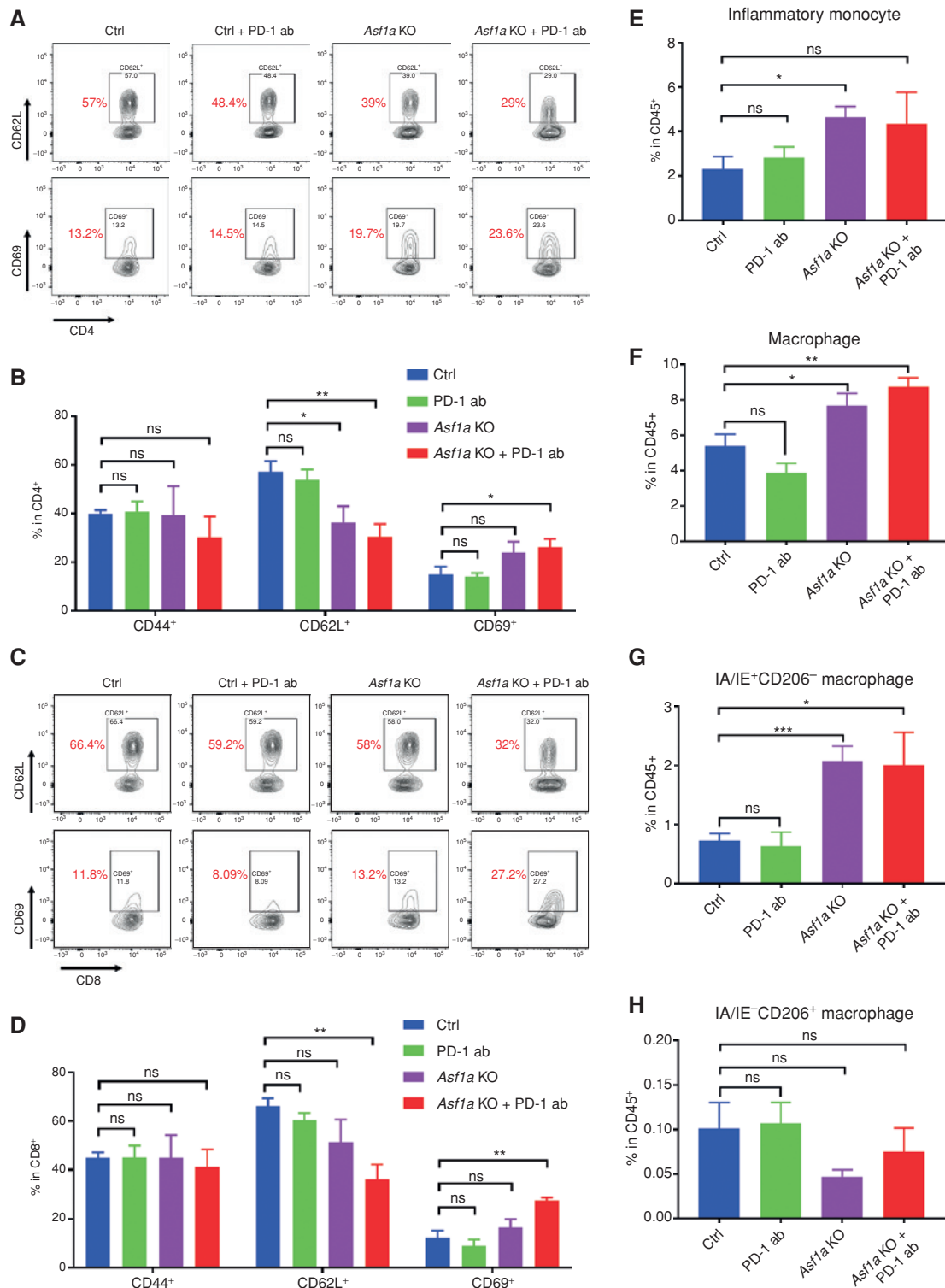
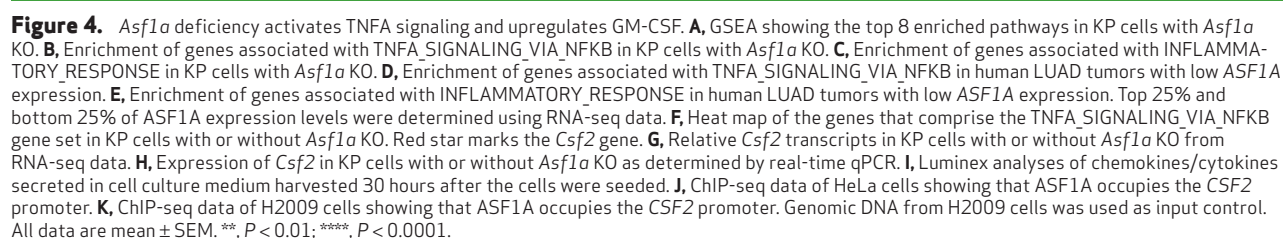


Figure 3. *Asf1a* deficiency and anti-PD-1 treatment promotes T-cell activation, inflammatory response, and M1-like macrophage polarization. **A**, Representative flow cytometry analysis of CD62L⁺ (naïve T-cell marker) and CD69⁺ (T-cell activation marker) populations of CD4⁺ T cells across all treatment groups. **B**, Bar graphs comparing the expression of CD44⁺ (T-cell activation marker), CD62L⁺ and CD69⁺ populations of CD4⁺ T cells across all treatment groups. **C**, Representative flow cytometry analysis of CD62L⁺ and CD69⁺ populations of CD8⁺ T cells. **D**, Bar graphs comparing the expression of CD44⁺, CD62L⁺, and CD69⁺ populations of CD8⁺ T cells. **E-H**, Flow cytometry analysis on changes in the expression of inflammatory monocytes (CD11B⁺/GR1⁻/SelectF⁻/Ly6C⁺; **E**), macrophages (CD11B⁺/GR1⁻/F4/80⁺; **F**), M1-like macrophages (CD11B⁺/GR1⁻/F4/80⁺/MHC-II⁺/CD206⁻; **G**), and M2-like macrophages (CD11B⁺/GR1⁻/F4/80⁺/MHC-II⁻/CD206⁺; **H**) in CD45⁺ cells. For all flow cytometry experiments, the whole tumor-bearing lungs from an intravenous injection model were harvested and processed after 1 week of treatment (Ctrl, *n* = 5; PD-1 ab, *n* = 5; *Asf1a* KO, *n* = 5; *Asf1a* KO + PD-1 ab, *n* = 5). All data are mean ± SEM. *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001.



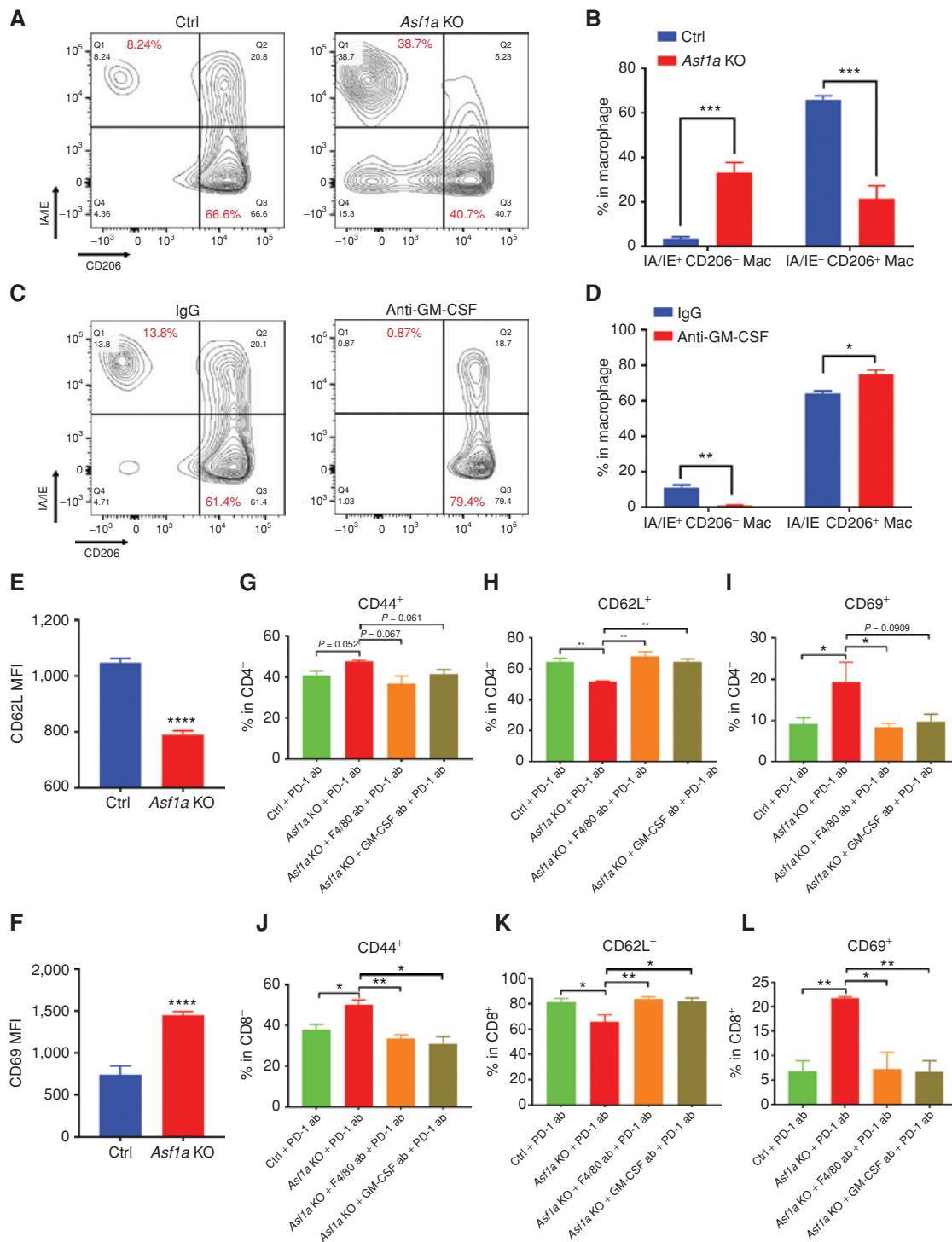


Figure 5. *Asf1a* deficiency promotes M1-like macrophage polarization and T-cell activation through upregulation of GM-CSF. **A**, Flow cytometry analysis of IA/IE and CD206 expression in macrophages cocultured with KP-ctrl cells or KP-*Asf1a* KO cells for 7 days. **B**, Bar graph showing the percentages of IA/IE⁺CD206⁻ (M1-like) macrophages and IA/IE⁻CD206⁺ (M2-like) macrophages from the coculture experiment shown in **A** (Ctrl, *n* = 4; *Asf1a* KO, *n* = 6). **C**, Flow cytometry analysis of IA/IE and CD206 expression in macrophages cocultured with KP cells in the presence of IgG or anti-GM-CSF for 7 days. **D**, Bar graph showing the percentages of IA/IE⁺CD206⁻ (M1-like) macrophages and IA/IE⁻CD206⁺ (M2-like) macrophages from the coculture experiment shown in **C** (IgG, *n* = 3; anti-GM-CSF, *n* = 3). **E–F**, Flow cytometry analysis of changes in expression of CD62L (**E**) and CD69 (**F**) in OT-I T cells cocultured with macrophages that were sorted from the coculture system shown in **A** (Ctrl, *n* = 3; *Asf1a* KO, *n* = 7). **G–L**, Flow analysis of CD44⁺ (**G**), CD62L⁺ (**H**), and CD69⁺ (**I**) populations in CD4⁺ T cells. **J–L**, Flow analysis of the expression of CD44⁺ (**J**), CD62L⁺ (**K**), and CD69⁺ (**L**) populations in CD8⁺ T cells. For flow cytometry analyses in **G–L**, whole tumor-bearing lungs from the transthoracic injection model were harvested and processed for flow cytometry analysis after 3 weeks of treatment (Ctrl + PD-1 ab, *n* = 4; *Asf1a* KO + PD-1 ab, *n* = 4; *Asf1a* KO + F4/80 ab + PD-1 ab, *n* = 5; *Asf1a* KO + GM-CSF ab + PD-1 ab, *n* = 4). All data are mean ± SEM. *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001; ****, *P* < 0.0001. MFI, mean fluorescence intensity.

macrophage polarization and promotes T-cell activation via upregulating GM-CSF.

To further evaluate the importance of GM-CSF-mediated M1 macrophage polarization in antitumor immunity, we applied F4/80 and GM-CSF neutralizing antibodies during the *in vivo* treatment study. F4/80 antibody blocked the *Asf1a* KO plus anti-PD-1 treatment efficacy in 2 mice (2/6), and GM-CSF antibody blocked the *Asf1a* KO plus anti-PD-1 treatment efficacy in 3 mice (3/6, 2 mice with high tumor burden died before week 3 MRI imaging; Supplementary Fig. S14A). We found F4/80 or GM-CSF blockade inhibited T-cell activation (Fig. 5G–L). Moreover, GM-CSF blockade also decreased the expression of M1 macrophage markers CD80, CD86, and IA/IE (Supplementary Fig. S14B–S14D). These data further support that GM-CSF upregulation-mediated M1 macrophage polarization plays pivotal roles in the anti-tumor immunotherapy.

Single-Cell Analysis of Intratumoral Inflammatory Cells Confirms That Combined Targeting of *Asf1a* and PD-1 Potentiates Macrophage and T-cell Activation

The recent development of single-cell genomics has provided a powerful tool to dissect transcriptomic heterogeneity in tumor-infiltrating immune cells (35, 36). Therefore, to provide a comprehensive and unbiased assessment of immunotherapeutic responses affected by *Asf1a* deficiency or/and anti-PD-1 treatment, we performed single-cell RNA-seq (scRNA-seq) on LUAD specimens. We harvested and pooled tumors from 2 mice in each group, and single suspension cells were collected and directly analyzed on the 10x Genomics platform. We identified distinct cancer cell, T-cell, natural killer (NK)-cell, B-cell, macrophage/monocyte, DC, and neutrophil clusters (Fig. 6A; Supplementary Fig. S15A and S15B). Consistent with our previous observations, there was

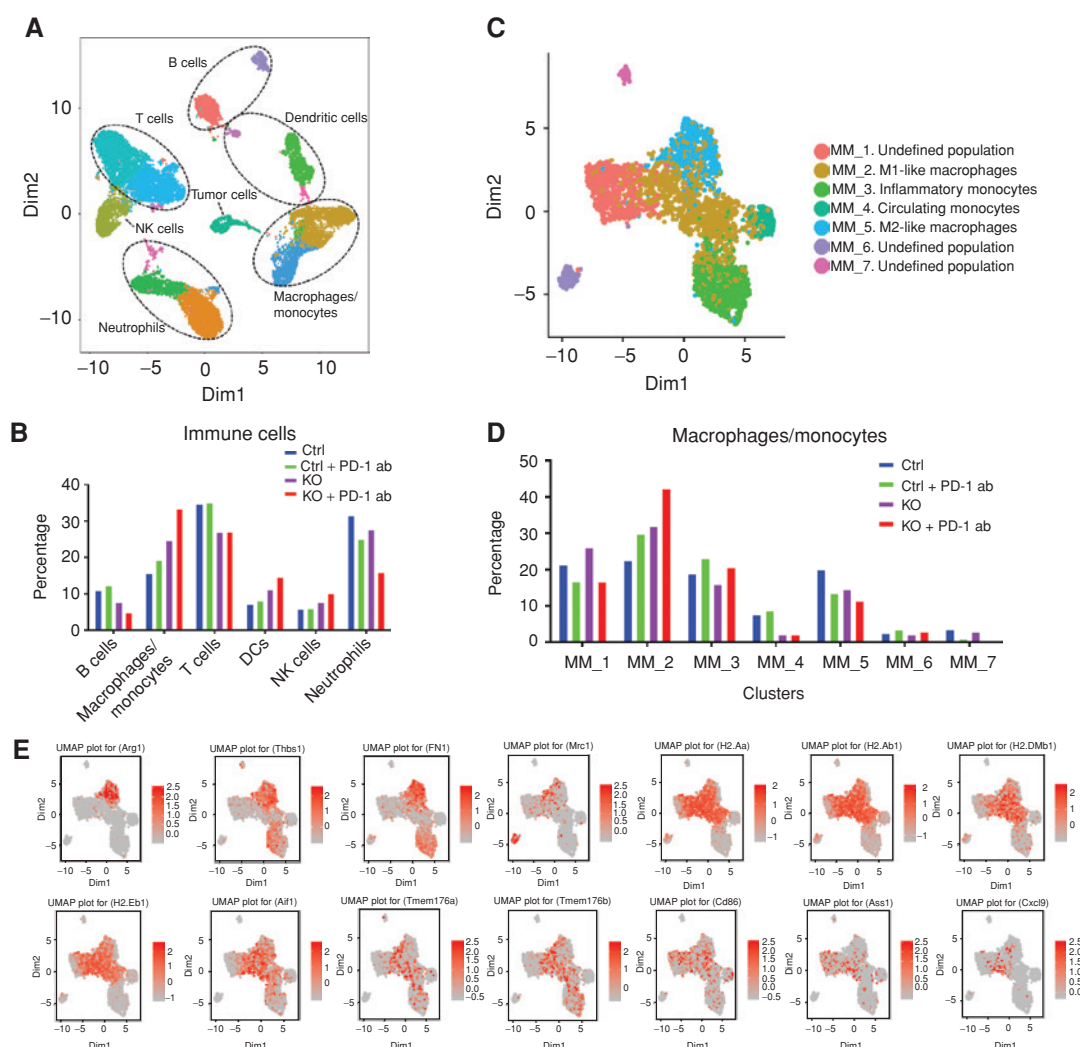


Figure 6. Single-cell analyses of intratumoral immune cell populations confirm the alterations of macrophage and T-cell populations. **A**, UMAP plot showing clusters of tumor cells (center) and intratumoral immune cell populations. **B**, Changes in the different immune compartments in response to indicated treatments. **C**, UMAP plot showing secondary clusters of macrophages/monocytes. **D**, Changes in different macrophage/monocyte subpopulations in response to indicated treatments. **E**, UMAP plots show the expression of M2 macrophage marker genes (*Arg1*, *Thbs1*, *Fnl1*, and *Mrc1*) and M1 macrophage marker genes (*H2.Aa*, *H2.Ab1*, *H2.DMb1*, *H2.Eb1*, *Aif1*, *Tmem176a*, *Tmem176b*, *Cd86*, *Ass1*, and *Cxcl9*) in the macrophage/monocyte subpopulations. (continued on next page)

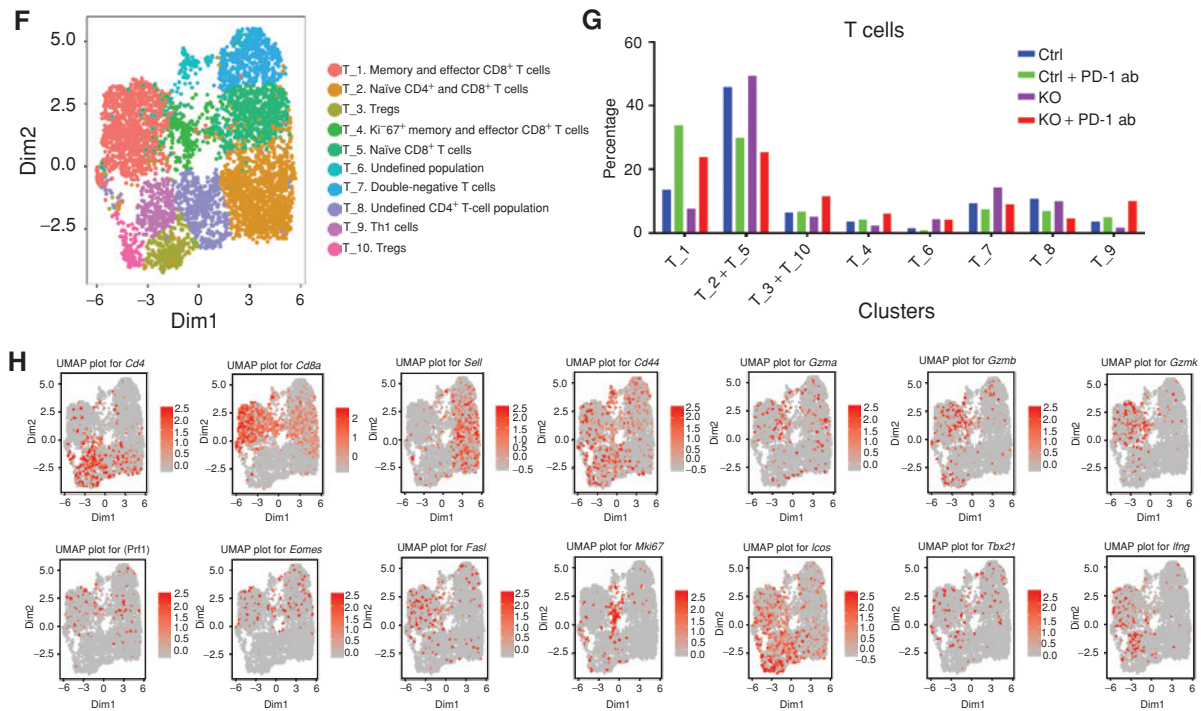


Figure 6. (Continued) F, UMAP plot showing secondary clusters of the T-cell population. G, Changes in different T-cell subpopulations in response to indicated treatments. H, UMAP plots showing the expression of T-cell marker genes (*Cd4*, *Cd8*, *Sell*, *Cd44*, *Gzma*, *Gzmb*, *Gzmk*, *Prf1*, *Eomes*, *Fas*, *Mki67*, *Icos*, *Tbx21*, and *Ifng*) in T-cell subpopulations.

an increase in the monocyte/macrophage cluster in tumors with *Asf1a* deficiency and/or anti-PD-1 treatment (Fig. 6B).

To further evaluate how *Asf1a* deficiency and anti-PD-1 immunotherapy affect the macrophage/monocyte transcriptome, we performed unbiased secondary clustering for these cells and identified 7 distinct subpopulations (Fig. 6C–E; Supplementary Fig. S16A and S16B). Cells in cluster MM_3 exhibited high *Itga4*, *Ly6c2*, *Ccr2*, and *Cd62l* (*Sell*) expression, but low *Cx3cr1* expression (Supplementary Fig. S16A), resembling classic inflammatory monocytes (37–39). Cells in cluster MM_4 exhibited low *Ly6c2*, *Cd62l* (*Sell*), and *Ccr2* expression, but high *Itga4* and *Cx3cr1* expression (Supplementary Fig. S16A), consistent with a nonclassic circulating monocyte phenotype (37–39). Cells in cluster MM_2 were characterized by the high expression of MHC-II genes (*H2.Aa*, *H2.Ab1*, *H2.DMb1*, and *H2.Eb1*), and high *Aif1*, *Tmem176a*, *Tmem176b*, *CD86*, *Ass1*, and *Cxcl9* expression (Fig. 6E), characteristic of M1-like macrophages. *Asf1a* deficiency and/or anti-PD-1 treatment markedly expanded the cluster 2 population. Cells in cluster MM_5 were defined by high expression of *Arg1*, *Thbs1*, *Fn1*, and *Cd206* (*Mrc1*; Fig. 6E), which was associated with immune-suppressive M2-like macrophages. Consistent with our previous observations, *Asf1a* deficiency and/or anti-PD-1 treatment reduced the cluster MM_5 population (Fig. 6D).

To assess how *Asf1a* deficiency and anti-PD-1 immunotherapy affect the transcriptome of tumor-infiltrating T cells, we also performed unbiased secondary clustering of the bulk T-cell population and identified 10 distinct subpopulations (Fig. 6F–H; Supplementary Fig. S17A and S17B). Cells in

clusters T_2 and T_5 were characterized by the high expression of CD4 or CD8, as well as the high expression of *Cd62l* (*Sell*), but low expression of *Cd44* (Fig. 6H), consistent with naïve T-cell phenotype. *Asf1a* deficiency and/or anti-PD-1 immunotherapy decreased T_2 and T_5 populations (Fig. 6G). Cells in cluster T_1 were characterized by high expression of cytotoxic markers including *Cd8*, *Gzma*, *Gzmb*, *Gzmk*, and *Prf1*, memory T-cell markers including *Eomes* and *Fas*, and T-cell activation markers including *Icos*, *Ifng*, *Ctla4*, *Lag3*, and *Pdcd1* (Fig. 6H; Supplementary Fig. S17A), indicating a heterogeneous population containing memory and effector CD8⁺ T cells. Anti-PD-1 immunotherapy increased the T_1 population (Fig. 6G). Cluster T_4 shared similar signature with cluster T_1, but was characterized by high expression of the gene encoding Ki-67 (*Mki67*) expression (Fig. 6H), indicative of highly proliferative CD8⁺ memory and effector T cells. *Asf1a* deficiency plus anti-PD-1 immunotherapy increased the T_4 population, which implies the memory and effector T cells started to expand from an early time point (7 days). Cells in cluster T_9 have high expression of *Cd4* and T-cell activation markers including *Icos*, *Ctla4*, *Tnfrsf4*, *Tnfrsf18*, *Tbet* (*Tbx21*), *Pdcd1*, *Tnfrsf18*, and *Tnfrsf4* (Fig. 6H; Supplementary Fig. S17A), indicative of Th1 cells. *Asf1a* deficiency plus anti-PD-1 immunotherapy increased the T_9 population (Fig. 6G), which is consistent with our previous observations (Supplementary Fig. S8L), supporting the enhanced M1 macrophage polarization. Cells in cluster T_3 and T_10 exhibited high expression of *Cd4*, *Foxp3*, and *Il2ra* (Supplementary Fig. S17A), characteristic of regulatory T cells (Treg). *Asf1a* deficiency plus anti-PD-1 immunotherapy

increased T₃ and T₁₀ populations (Fig. 6G), possibly suggesting an acute positive feedback to a potentiated or activated tumor immune environment by short-term treatment.

In summary, these results support the notion that *Asf1a* deficiency and anti-PD-1 combination therapy restrains tumor progression through promoting inflammatory M1-like macrophage polarization and T-cell activation.

DISCUSSION

Compared with *in vitro* CRISPR screens, *in vivo* models are the more physiologically relevant systems to screen for new immunotherapy targets. Particularly, using small focused libraries is a practical strategy for *in vivo* CRISPR screens. Manguso and colleagues first used an *in vivo* CRISPR screen using an sgRNA sublibrary containing ~2,000 selected genes to identify novel immunotherapy targets in melanoma (18). Our current study is the first *in vivo* CRISPR screen to identify novel immunotherapy targets in lung cancer utilizing an epigenetic sgRNA library containing 524 epigenetic genes. Internal quality controls indicated that our *in vivo* CRISPR screen functioned successfully, and our study provided further proof of principle for the utilization of *in vivo* CRISPR screens for detecting potential targets for cancer immunotherapy.

The development of immune-checkpoint targeted antibodies has brought hope to some patients with advanced-stage *KRAS*-mutant LUAD, but the majority of patients remain unresponsive to immunotherapy. The modest response rates highlight an urgent need for new therapeutic approaches to augment the antitumor immune response. Our *in vivo* epigenetic CRISPR screen identified *Asf1a* as an immunotherapeutic target whose inhibition synergizes with anti-PD-1 treatment by promoting M1-like macrophage differentiation and enhancing T-cell activation. Our results thus provide a rationale for combination therapy consisting of ASF1A inhibition and anti-PD-1 immunotherapy for patients with LUAD. Unfortunately, we could not access any human lung cancer patient data with immune-checkpoint blockade treatment to analyze whether ASF1A loss is a biomarker for immune-checkpoint blockade efficacy. It would be very interesting if we could use this finding to strategize treatment of patients with cancer based on their ASF1A expression levels, and to see whether ASF1A-low patients treated with anti-PD-1 antibody will have better outcome and survival.

ASF1A plays a role in regulating gene transcription (23), but its function in cancer has been only scarcely studied. Individual reports showed that ASF1A inhibition elicits DNA damage in cancer cells (28). Although emerging evidence indicates that accumulated DNA damage may lead to increased inflammation through the cGAS-cGAMP-STING pathway (40), we found no significant difference in γ H2AX levels in KP cells with or without *Asf1a* knockout, indicating that the enhanced inflammation in *Asf1a* knockout tumors might not be due to the DNA-damage response. Transcriptional profiling revealed that *Asf1a* depletion led to the upregulation of a variety of factors, including GM-CSF, which was confirmed in cell culture supernatants. Furthermore, the ChIP-seq data confirmed the occupancy of ASF1A on the promoter of *CSF2*. GM-CSF (encoded by *CSF2*) promotes M1-like macrophage

differentiation (31, 32). Thus, our data reveal a mechanism for the increased abundance of M1-like macrophages in *Asf1a*-deficient tumors.

The function of GM-CSF in antitumor immunity is controversial. GM-CSF plays an important role in DC development (41), and DCs are critical in T-cell priming. Moreover, GM-CSF was reported to promote M1 macrophage polarization (42), and M1 macrophages promote antitumor immunity. By contrast, tumor-derived GM-CSF is necessary and sufficient to drive the development of CD11B⁺GR1⁺ MDSCs that suppressed antigen-specific T cells, and thus promotes tumor progression (43, 44). However, in our model, although *Asf1a* KO-induced elevation of GM-CSF expression did not increase the MDSC population in the tumor microenvironment (Supplementary Fig. S10C), it is still unclear whether overall GM-CSF blockade will inhibit the suppressive function of MDSCs. It is hard to segregate tumor-derived GM-CSF versus overall GM-CSF systemically in the current study. Moreover, GM-CSF works in concert with other chemokines/cytokines to promote the suppressive function of MDSCs (45), and the alterations of other chemokines/cytokines may also be involved in the regulation of the MDSC population in our model.

Elevated expression of MHC-II in tumor-associated macrophages promotes antigen presentation and T-cell priming (14). In line with this, our data revealed an upregulation of MHC-II in tumor-associated macrophages (increased M1-like macrophage) in *Asf1a*-deficient tumors, which could potentially enhance T-cell activation and augment sensitivity to anti-PD-1 immunotherapy.

Interestingly, *Asf1a* deficiency alone did not robustly reduce tumor progression in our KP orthotopic allograft model, suggesting that increased M1-like macrophage differentiation may not be sufficient to achieve a sustained antitumor immune response. However, tumor cell-intrinsic *Asf1a* deficiency coupled with anti-PD-1 treatment substantially increased the antitumor immune response. In *Asf1a*-deficient tumors treated with anti-PD-1, we observed a decrease in CD62L expression coupled with an increase in CD69, 4-1BB, OX40, and ICOS expression on intratumoral T cells, suggesting the enhanced T-cell activation in this context. These data suggest that combined targeting of *Asf1a* and PD-1 may enable the efficacy for targeting additional checkpoint or costimulatory receptors.

Our study focused on the macrophage population, but we could not exclude the possibility that the other immune-cell populations, such as DCs, neutrophils, and NK cells, may also play important roles in the antitumor immune response in *Asf1a*-deficient tumors treated with anti-PD-1. scRNA-seq analyses of intratumoral immune-cell populations revealed an increase in DCs and NK cells and decrease in neutrophils. GM-CSF promotes inflammatory monocyte differentiation into DCs (41), which play critical roles in antigen presentation and T-cell priming (46). NK cells also play important roles in the antitumor immune response (47). By contrast, TANs inhibit an antitumor immune response (48), and decreased neutrophil levels support enhanced antitumor immune responses. Thus, the mechanism of the enhanced antitumor immune response in *Asf1a*-deficient tumors treated with anti-PD-1 may involve multiple innate immune populations. Future work is needed

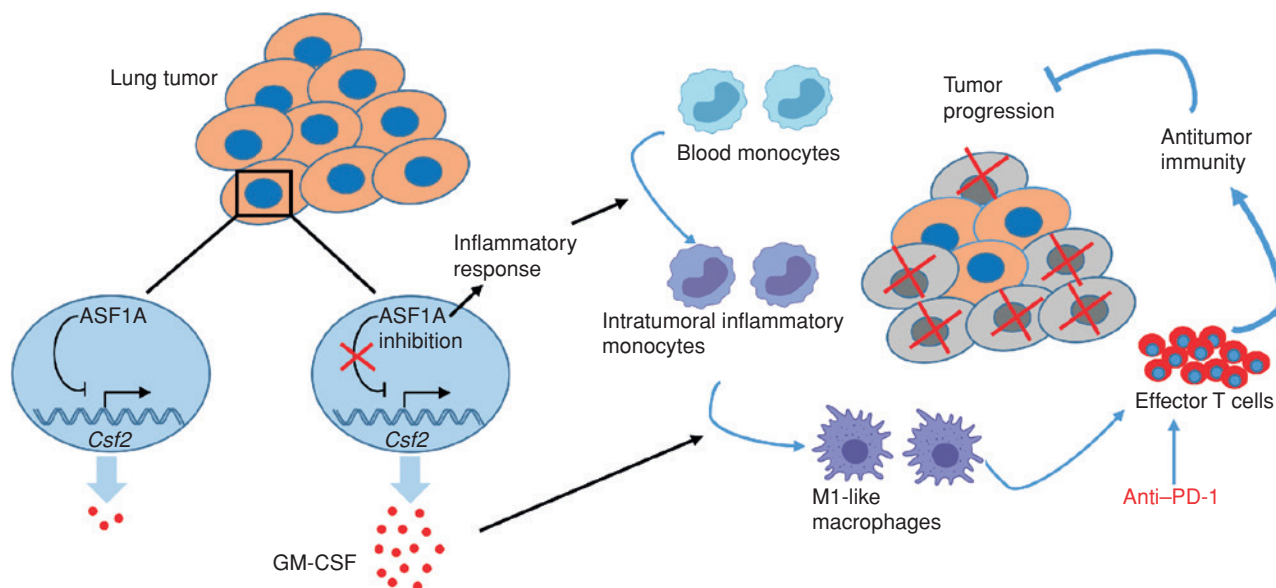


Figure 7. Working model for *Asf1a* deficiency combined with anti-PD-1 combination therapy. Tumor cell-intrinsic *Asf1a* deficiency promotes an inflammatory response and GM-CSF secretion, which promotes M1-like macrophage polarization and T-cell activation. Anti-PD-1 therapy also promotes T-cell activation. Thus, *Asf1a* KO synergizes with anti-PD-1 treatment to promote antitumor immunity.

to address whether *Asf1a* can directly affect the activity of immune cells.

The development of a specific ASF1A inhibitor will potentially facilitate the translational significance of current work. Seol and colleagues developed multiple ASF1A inhibitors and tested them *in vitro* (49). However, a high concentration (30–50 $\mu\text{mol/L}$) of these inhibitors was required to interrupt the interaction between ASF1A and histone H3. In HeLa cells, only a high-concentration (40 $\mu\text{mol/L}$) treatment with the putative inhibitors could inhibit H3K56 acetylation. These compounds were not fully optimized for targeting ASF1A, and there are no data from *in vivo* studies to support their targeting efficacy. Further effort is needed to optimize the drug with lower IC_{50} and better pharmacodynamic efficacy for *in vivo* testing and clinical applications.

ASF1A inhibition may cause cell-cycle arrest in some cell types (28). As in the case of CDK4/6 inhibition, despite the fact that CDK4/6 inhibitors can cause cell-cycle arrest even in some normal cell types and immune cells, they were still approved by the FDA for the treatment of certain patients with breast cancer (50). Besides causing cell-cycle arrest, CDK4/6 inhibitors could also affect other aspects of cancer cell behavior such as enhancing antitumor immunity if dosing schedule is optimized as intermittent to bypass the cell-cycle arrest (51, 52), suggesting that CDK4/6 is still an attractive therapeutic target. Similarly, beyond the possible cell-cycle arrest in some cell types (28), here we showed that ASF1A is a potential target and its inhibition can synergize with anti-PD-1 treatment to restrain cancer development with enhance antitumor immunity through both macrophage and T cells. There is a need to develop more efficient and selective ASF1A inhibitors for further preclinical and clinical evaluation of the feasibility of targeting ASF1A in lung cancer and other cancers. Upon drug

optimization, further investigations of the treatment strategy are needed to minimize the cell-cycle arrest side effect of ASF1A inhibition.

In summary, we performed an *in vivo* epigenome-focused CRISPR screen and identified ASF1A as a promising candidate in sensitizing immunotherapy. Functional and mechanistic studies showed that tumor cell-intrinsic *Asf1a* deficiency promotes inflammation and M1-like macrophage polarization and further promotes T-cell activation in combination with anti-PD-1 treatment (Fig. 7). ASF1A loss combined with anti-PD-1 treatment exerted significant inhibition of tumor growth. Thus, combining ASF1A inhibition with immune-checkpoint blockade such as anti-PD-1 treatment might serve as a potential novel immunotherapy strategy for patients with lung cancer.

METHODS

Cell Culture, Plasmid Construction, and Lentivirus Infection

HEK-293T cells and the colon cancer cell line MC38 were cultured in Dulbecco's Modified Eagle Medium (Gibco) with 10% fetal bovine serum (FBS). Mouse LUAD lines KP and KP-2 (C57BL/6 background), and human LUAD cell line NCI-H2009 were cultured in RPMI-1640 (Gibco) with 10% FBS. All cell lines used in this study were tested as *Mycoplasma*-negative using the Universal Mycoplasma Detection Kit (ATCC 30-1012K). Plasmids pLenti-Cas9-Puro, pXPR-GFP-Blast, pLKO.1-Tet-on, PSPAX2, and PMD2.G were purchased from Addgene.

The sgRNAs specific for mouse *Asf1a* were cloned into pXPR-GFP-Blast vector using the Gibson Assembly Kit (E2611L, NEB). The target sequences are as follows:

sg*Asf1a*-1: 5'-CTGATTACTTGACCTACCG-3'
 sg*Asf1a*-2: 5'-TCTGGGATGAGTCCTGCATT-3'
 sg*Asf1a*-3: 5'-GATCACCTTCGAGTGCATCG-3'
 sg*Asf1a*-4: 5'-TAGGCTGATGCACCGAATGC-3'

The shRNAs specific for mouse *Asf1a* were cloned into pLKO.1-Tet-on vector with the AgeI/EcoRI sites. The target sequences are as follows:

sh*Asf1a*-2: 5'-CTAAGCTTCAAAGGAATATTT-3'
sh*Asf1a*-3: 5'-TGAGCAAATTGTGGATTATAA-3'

The shRNAs specific for human *ASF1A* (pLKO.1 vector) were purchased from Sigma. The target sequences are as follows:

sh*ASF1A*-1: 5'-GCCAGATGTTAACTTTGAAT-3' (TRCN0000074268)
sh*ASF1A*-4: 5'-AGGCGTAACTGTTGTGCTAAT-3' (TRCN0000074271)

To generate lentivirus, HEK-293T cells were cotransfected with pLenti-Cas9, pXPR-GFP-sgRNA-Blast, or pLKO.1-Tet-on-shRNA plasmid, and packaging plasmids PSPAX2 and PMD2.G using Lipofectamine 3000 (Invitrogen). Viral particles released into the cell culture supernatant were filtered with 0.45- μ m filters (Corning) to remove cellular debris. KP cells were transduced by culturing with viral supernatants in the presence of Polybrene (Sigma) to increase infection efficiency. Stable cell lines were selected and maintained in cell culture media containing 2 μ g/mL puromycin or 5 μ g/mL blasticidin.

Colony Formation Assay

Cells were trypsinized to produce a single-cell suspension. One thousand cells were counted and plated in each well of a 6-well plate. Medium was changed every 2 days. After 9 days, cells were fixed with 70% ethanol for 10 minutes, and the cells were stained with 0.5% crystal violet (dissolved in 20% methanol) for 5 minutes and washed. Photos were taken and quantified using ImageJ.

Construction of an Epigenetic-Focused sgRNA Library

We obtained sgRNA oligo pools from the Belfer Center for Applied Cancer Science at the Dana-Farber Cancer Institute (53). The library contains 7,780 sgRNAs, including sgRNAs targeting 524 epigenetic regulators, 173 control genes (essential genes, immune modulators, etc.), and 723 nontargeting sgRNAs. For each gene, there are 8 to 12 sgRNAs. Additional details of the library are included in Supplementary Table S1. The sgRNA library was inserted into the pXPR-GFP-Blast vector using the Gibson Assembly Kit (NEB), expanded by transformation into electrocompetent cells (Invitrogen) by electroporation. Library representation was maintained at least 1,000 $\times$ at each step of the preparation process.

In Vivo Screen Using a KP-Cas9 Lung Cancer Cell Line in B6 Mice

We isolated single clones from a KP lung cancer cell line with pure C57BL/6 background. To determine how propagation as a tumor *in vivo* affects basal sgRNA library distribution, clones without Cas9 expression were transduced at a low multiplicity of infection (MOI) with the sgRNA library for use as barcodes. Following several days of expansion in medium with 5 μ g/mL blasticidin, a pellet of at least 8 million cells was kept to evaluate the initial sgRNA distribution, and the remainder was injected subcutaneously into B6-*Rag1*^{-/-} and B6 WT mice. Tumors were allowed to grow to 500 mm³, then excised for NGS analysis to evaluate the final library distribution. For the functional screen, KP-Cas9 cells were generated by viral transduction and Cas9 expression was confirmed by Western blot. Cas9-mediated DNA editing was confirmed using sgRNAs targeting the essential gene *Rbx1*. KP-Cas9 clones with validated Cas9 activity were transduced at a MOI of 0.2 with lentivirus produced from the libraries with at least 1,000-fold coverage (cells per construct) in each infection replicate. Transduced KP cells were expanded *in vitro* for 2 weeks and then subcutaneously implanted into B6-*Rag1*^{-/-} mice and C57BL/6 mice. These mice were then treated with anti-PD-1 or isotype control (3 times per week) on day 7 when the average tumor size reached 60 mm³; tumors were harvested on day 24 when the

tumor size was roughly 500 mm³. To better keep the *in vivo* sgRNA representations, there were 12 tumors from 6 mice in each group. Genomic DNA of tumors or cells was extracted using the DNA Blood Midi Kit (Qiagen). PCR was used to amplify the sgRNA cassette, and NGS was performed on an Illumina HiSeq to determine sgRNA abundance.

Data Analysis for CRISPR Screen

Adaptor sequences were trimmed using cutadapt (v1.18), and untrimmed reads were removed. Then, sequences after the 20-base gRNAs were cut using fastx-toolkit (v0.0.13; http://hannonlab.cshl.edu/fastx_toolkit/index.html), gRNAs were mapped to the annotation file (0 mismatch), and read count tables were made. The count tables were normalized based on their library size factors using DESeq2 (54), and differential expression analysis was performed. Further, MAGeCK (0.5.8; ref. 55) was used to normalize the count table based on median normalization and fold changes, and significance of changes in the conditions was calculated for genes and sgRNAs. Pathway analysis and GSEA were performed using clusterProfiler R package (v3.6.0; ref. 56). All downstream statistical analyses and generating plots were performed in R (v3.1.1; <http://www.r-project.org>).

Animal Studies

All mouse work was reviewed and approved by the Institutional Animal Care and Use Committee at either NYU School of Medicine or Dana-Farber Cancer Institute. Specific pathogen-free facilities were used for housing and care of all mice. Six-week-old male B6-*Rag1*^{-/-} and B6 WT mice were purchased from The Jackson Laboratories. For screen, 8 $\times$ 10⁶ library-transduced KP-Cas9 cells were resuspended in phosphate-buffered saline (PBS) and subcutaneously inoculated into the flanks of B6 mice. Mouse anti-PD-1 ab (29F.1A12; ref. 57) or isotype control was administered 3 times per week (Monday, Wednesday, and Friday) at 200 μ g/mouse via intraperitoneal injection. Tumor size was measured every 3 days using calipers to collect maximal tumor length and width. Tumor volume was estimated with the following formula: ($L \times W^2$)/2.

For the single-nodule orthotopic lung cancer model, 0.25 million cells in 20 μ L PBS were injected into the left lung through the ultrasound guided transthoracic injection. Tail-vein injection was also used here as an alternative orthotopic model, and 1 million cells were injected into each mouse. MRI was used to monitor tumor formation and progression in orthotopic models. Randomization of mouse groups was performed when appropriate. Mice were treated with isotype control, anti-PD-1 antibody (29F.1A12; ref. 57). For F4/80 or GM-CSF blocking study, mice were injected intraperitoneally with either anti-F4/80 antibody (200 μ g/mouse, clone CI:A3-1, BioXCell) or anti-GM-CSF antibody (200 μ g/mouse, clone MP1-22E9, BioXCell) 48 and 24 hours before the beginning of anti-PD-1 treatment of KP-*Asf1a* KO tumors, and 3 times per week thereafter. CO₂ inhalation was used to euthanize mice when we harvested the tissues.

MRI Quantification

Mice were anesthetized with isoflurane to perform lung MRI using the BioSpec USR70/30 horizontal bore system (Bruker) to scan 24 consecutive sections. Tumor volume within the whole lung was quantified using 3-D slicer software to reconstruct MRI volumetric measurements, as described previously (58). Acquisition of the MRI signal was adapted according to cardiac and respiratory cycles to minimize motion effects during imaging.

RNA-seq and Data Analyses

RNA-seq on KP cells with or without *Asf1a* KO was performed in NYU Langone Health Genome Technology Core. STAR 2.4.2a (59) was

used to align the RNA-seq samples to the reference mouse genome (mm9) and to count the number of reads mapping to each gene in the Ensembl GRCm38.80 gene model. Differential expression between the different groups was performed through the use of DESeq2 (60). Differential expression analysis was done using R (v.3.5.1; <http://www.R-project.org/>) and the DESeq2 package (v.1.10.0).

Gene set enrichment analysis was done using GSEA (v.3.0) and gene sets from MSigDB (v.5.0). We used the “preranked” algorithm to analyze gene lists ranked by the negative decadic logarithm of *P* values multiplied by the value of $\log_2FC$ obtained from the differential expression analysis with DESeq2.

TCGA RNA-seq Data Analysis

Level 3 RNA-seq data for TCGA LUADs were obtained through the TCGA portal. Data were sorted based on the expression level of *ASF1A*, and the samples were separated into quarters. The top 25% expression group (high expression) was compared with the low 25% expression group (low expression) by GSEA as outlined in the RNA-seq data analysis section. The gene list for GSEA input was ranked by the value of $\log_2FC$, where FC is defined by the ratio of low expression:high expression.

ChIP-seq and ATAC-seq

ChIP was performed in the human lung cancer cell line NCI-H2009 using the Pierce Magnetic ChIP Kit (26157, Thermo Fisher Scientific) following the manufacturer's instructions. Antibody against ASF1A (2990s, Cell Signaling Technology) was used. ChIP DNA was purified and sent to NYU Langone Health Genome Technology Center for library construction and sequencing.

For ATAC-seq, freshly harvested cells were directly sent to NYU Langone Health Genome Technology Center for library construction and sequencing.

ChIP-seq and ATAC-seq Data Analysis

All of the reads from the sequencing experiment were mapped to the reference genome using the Bowtie 2 (v2.2.4; ref. 61), and duplicate reads were removed using Picard tools (v.1.126; <http://broadinstitute.github.io/picard/>). Low-quality mapped reads (MQ < 20) were removed from analysis. The reads per million normalized BigWig files were generated using BEDTools (v.2.17.0; ref. 62) and the bedGraphToBigWig tool (v.4). Peak calling was performed using MACS (v1.4.2; ref. 63) and peak count tables were created using BEDTools. Differential peak analysis was performed using DESeq2 (54). ChIPseeker (v1.8.0; ref. 64) R package was used for peak annotations, and motif discovery was performed using HOMER (v4.10; ref. 65). ngs.plot (v2.47; ref. 65) and ChIPseeker were used for transcription start site visualizations and quality controls. Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis and Gene Ontology analysis were performed using the clusterProfiler R package (v3.0.0; ref. 56). To compare the level of similarity among the samples and their replicates, we used two methods: principal component analysis and Euclidean distance-based sample clustering. Downstream statistical analyses and generating plots were performed in R environment (v3.1.1; <http://www.R-project.org/>).

Tumor-Infiltrating Immune Cell Isolation and FACS Analysis

Mice were euthanized, and lungs were perfused with sterile PBS through heart perfusion from the left ventricle after collection of bronchoalveolar lavage (BAL) fluid. Whole lung was minced and digested in collagenase D (11088866001, Roche) and DNase I (10104159001, Roche) in Hank's Balanced Salt Solution at 37°C for 30 minutes. After incubation, digested tissue was filtered through a 70- μ m cell strainer (Thermo Fisher Scientific) to obtain

single-cell suspensions. Separated cells were treated with $1 \times$ RBC lysis buffer (BioLegend) to lyse red blood cells. Live cells were determined with a LIVE/DEAD Fixable Aqua Dead Cell Stain Kit (Molecular Probes). Cell pellets were resuspended in PBS with 2% FBS for FACS analysis. Cells were stained with the indicated cell-surface markers and fixed/permeabilized using a Fixation/Permeabilization Kit (eBioscience). Cells were imaged on a BD Biosciences LSRFortessa and analyzed with FlowJo software. The gating strategy was described previously (66).

Flow Antibodies

Lung-infiltrating immune cells were stained with fluorochrome-coupled antibodies against mouse CD45 (clone 30-F11, BioLegend), CD3 (clone 17A2, BioLegend), CD4 (clone GK1.5, BioLegend), CD8 (clone 53-6.7, BioLegend), CD44 (clone IM7, BioLegend), CD62L (clone MEL-14, BioLegend), CD69 (clone H1.2F3, BioLegend), 4-1BB (CD137, clone 17B5, BioLegend), GITR (CD357, clone DTA-1, BioLegend), GZMB (clone GB11, BD Horizon), OX40 (CD134, clone OX-86, BioLegend), ICOS (clone 7E.17G9, BD OptiBuild), CD11B (clone M1/70, BioLegend), CD11C (clone N418, BioLegend), Ly6G (clone 1A8, BioLegend), Siglec F (clone E50-2440, BD Pharmingen), Ly6C (clone HK1.4, BioLegend), GR1 (Ly6G/Ly6C, clone RB6-8C5, BioLegend), CD103 (clone 2E7, BioLegend), F4/80 (clone BM8, BioLegend), CD80 (clone 16-10A1, BioLegend), CD86 (clone GL-1, BioLegend), IA/IE (clone M5/114.15.2, BioLegend), and CD206 (clone C068C2, BioLegend).

Single-Cell RNA-seq

Single-cell suspensions were achieved as described above and sorted using DAPI staining. Cells were then resuspended into single cells at a concentration of 1×10^6 per mL in $1 \times$ PBS with 0.4% BSA for 10x genomics processing. Cell suspensions were loaded onto a 10x Genomics Chromium instrument to generate single-cell gel beads in emulsion (GEM). Approximately 5,000 to 10,000 cells were loaded per channel. scRNA-seq libraries were prepared using the following Single Cell 3' Reagent Kits: Chromium Single Cell 3' Library and Gel Bead Kit v2 (PN-120237), Single Cell 3' Chip Kit v2 (PN-120236), and i7 Multiplex Kit (PN-120262; 10x Genomics) as previously described (67), and following the Single Cell 3' Reagent Kits v2 User Guide (Manual Part # CG00052 Rev A). Libraries were run on an Illumina HiSeq 4000 system (SY-401-4001, Illumina) as 2×150 paired-end reads, one full lane per sample, for approximately >90% sequencing saturation.

Single-Cell RNA-seq Data Analysis

After confirming the integrity of the cDNA, quality of the libraries, number of cells sequenced, and mean number of reads per cell, as a quality control, we used the Cell Ranger package to map the reads and generate gene-cell matrices. A quality control was then performed on the cells to calculate the number of genes, UMIs, and the proportion of mitochondrial genes for each cell using iCellR R package (v0.99.0; <https://github.com/rezakji/iCellR>), and the cells with a low number of covered genes (gene count < 500) and high mitochondrial counts (mt-genes > 0.1) were filtered out. Following this, the matrix was normalized based on ranked geometric library size factor (ranked glsf) using iCellR. Geometric library size factor normalization is a common normalization method used by popular tools such as DESeq2 (54); however, here we used only the top-ranked genes (top 500 genes sorted by base mean). This is for reducing the effect of dropouts (nonzero events counted as zero) in normalization by taking into account only the highly expressed genes. A general gene statistics analysis was then performed to calculate gene dispersion, base mean, and cell coverage to use to build a gene model for performing principal component analysis (PCA). Genes with high coverage (top 500) and high dispersion (dispersion > 1.5) were chosen, and PCA was performed; a second round of PCA was

performed based on the top 20 and bottom 20 genes predicted in the first 10 dimensions of PCA to fine-tune the results, and clustering was performed (iCellR options; clust.method = "kmeans," dist.method = "euclidean," index.method = "silhouette") on principal component with high standard deviation (top 10 PC), and *t*-distributed stochastic neighbor embedding (t-SNE) was performed. Uniform Manifold Approximation and Projection (UMAP) was also performed on the top 10 PCs. Marker genes for each cluster were determined based on fold change and adjusted *P* value (*t* test), and average gene expression for each cluster was calculated using iCellR. Marker genes were visualized on heat maps; bar plots and box plots for each cluster were used to determine the cell types using the ImmGen database (<https://www.immgen.org/>).

Western Blots and Antibodies

Cells were lysed in RIPA buffer (Pierce) containing protease/phosphatase inhibitor cocktail (Thermo Scientific). Protein concentration was measured using the BCA assay (Pierce). Equivalent amounts of each sample were loaded on 4% to 12% Bis-Tris gels (Invitrogen), transferred to nitrocellulose membranes, and immunoblotted with antibodies directed against Cas9 (MA1-202, Thermo Fisher), ASF1A (2990s, Cell Signaling Technology), ASF1B (PA5-67639, Invitrogen), γ H2AX (9718s, CST), and β -actin (Ab8227, Abcam). IRDye 800-labeled goat anti-rabbit IgG and IRDye 680-labeled goat anti-mouse IgG secondary antibodies were purchased from LI-COR Biosciences, and membranes were detected with an Odyssey detection system (LI-COR Biosciences).

Luminex Analyses

Cell culture medium was harvested 30 hours after seeding cells. MDOTS was performed as previously described (68), and culture medium was harvested 3 days after treatment. Cytokine/chemokine profiling of cell culture medium was performed using a Milliplex MAP Kit (Millipore) and measured on Multiplex Analyzer (Millipore). Concentrations (pg/mL) of each protein were derived from 5-parameter curve fitting models. Fold changes relative to the control were calculated and plotted as log₂ fold change. Lower and upper limits of quantitation (LLOQ/ULOQ) were derived from standard curves for cytokines above or below detection.

In Vitro Coculture Assay

Bone marrow-derived macrophages (BMDM) were prepared and cultured as previously described (30). In selected experiments, tumor cells and bone marrow cells were mixed and plated in a well of a 6-well plate at a ratio of 2:3 (0.4 million:0.6 million). Cells were grown in RPMI-1640 with 10% FBS and 1 ng/mL macrophage colony-stimulating factor, and medium was changed every 2 days. After 7 days of coculture, cells were harvested for immunostaining and FACS analyses.

For T-cell activation experiment, GFP⁺/DAPI⁺/CD11b⁺/GR1⁺/F4/80⁺ macrophage cells from the above coculture system were sorted, stimulated with OVA 257-264 peptide (10 μ g/mL, 1 hour), and then further cocultured with OT-I T cells in a 1:5 ratio (0.025 million macrophages:0.125 million T cells) in a 96-well U bottom plate, and T-cell activation medium was used. After 5 days of coculture, OT-I T cells were directly used for immunostaining and flow cytometry analysis.

B6 WT mice were intravenously injected with KP-Ctrl or KP-Asf1a KO cells, and tumor formation in the lung was indicated by MRI imaging. These mice were treated with PD-1 ab for 1 week (3 times per week). Pan T cells from the spleens or lungs of these pretreated mice were isolated using the Pan T-cell Isolation Kit (mouse, 130-095-130), and cocultured with KP-Ctrl tumor cells in a 1:3 ratio (0.12 million T cells:0.36 million tumor cells). After 2 days of coculture, Cell Counting Kit-8 (ALX-850-039-KI02, Enzo) was used to measure tumor cell activity.

Rechallenge Experiment

One million MC38-Ctrl or MC38-Asf1a KO cells were subcutaneously injected to the left flanks of mice. When the average tumor size reached 120 mm³, mice were treated with anti-PD-1 for 2 weeks. Following treatment, 1 million MC38-Ctrl cells were subcutaneously injected to the right flanks of pretreated mice bearing MC38-Ctrl tumors or MC38-Asf1a KO tumors. After 8 days, tumors on the right flanks were measured and harvested for further analysis.

Quantitative RT-PCR

Total RNA was extracted from cells using an RNeasy Plus Mini Kit (Qiagen), and cDNA was generated with a High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems). Quantitative PCR was performed using SYBR Green PCR Master Mix (Applied Biosystems), and transcript levels were normalized to the internal control, actin/ACTIN. Samples were run in triplicate. The primer sequences are as follows:

mAsf1a-F: 5'-ATGTGGGCTCTGCAGAAAGT-3'
mAsf1a-R: 5'-CTGTTACCCCACTGCATCT-3'
hASF1A-F: 5'-CCTTTCTACAACCCGTTCCA-3'
hASF1A-R: 5'-ACTTTCTGCAGAGCCACAT-3'
mCsf2-F: 5'-ATGCCTGTCACGTTGAATGA-3'
mCsf2-R: 5'-CCGTAGACCTGCTCGAATA-3'
hCSF2-F: 5'-TTCTGCTGTGCATCCCTTT-3'
hCSF2-R: 5'-CTTGGTCCCTCAAGATGAC-3'
mActin-F: 5'-CTGTCCCTGTATGCCTCTG-3'
mActin-R: 5'-ATGTCACGCACGATTTC-3'
hACTIN-F: 5'-CACCATTGGCAATGAGCGGTTTC-3'
hACTIN-R: 5'-AGGTCTTTGCGGATGTCCACGT-3'

Statistical Analysis

All statistical analyses were carried out using GraphPad Prism 7. Data were analyzed by Student *t* test (two-tailed). Survival was measured according to the Kaplan–Meier method and analyzed by log-rank (Mantel–Cox) test. *P* < 0.05 was considered significant. Error bars represent standard error of the mean (SEM).

Data Access

NGS data for CRISPR screen, RNA-seq data, scRNA-seq data, ChIP-seq data, and ATAC-seq data have been deposited in the National Center for Biotechnology Information's Gene Expression Omnibus and are accessible through GEO Series accession numbers GSE127205 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE127205>), GSE127232 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE127232>), GSE133604 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE133604>), and GSE138571 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE138571>).

Disclosure of Potential Conflicts of Interest

V. Velcheti is a consultant for Foundation Medicine, AstraZeneca, Genentech, BTG Foundation, Novartis, Bristol-Myers Squibb, and Merck. G.J. Freeman is a consultant for Roche, Bristol-Myers Squibb, iTeos, Elpiscience, Monopteros, Xios, Origimed, Nextpoint, Novartis, Surface Oncology, Elstar, SQZ, and Adaptimmune, reports receiving commercial research grants from Novartis, Roche, UCB, Ipsen, and Quark, and has ownership interest (including patents) in Nextpoint, Xios, Dako, Novartis, Triursus, iTeos, Roche, Merck, Bristol-Myers Squibb, EMD-Serono, Boehringer Ingelheim, and AstraZeneca. J. Qi is a consultant for Epiphany. G. Miller reports receiving commercial research support from GSK, Puretech, and Pfizer. K.-K. Wong is a consultant for G1 Therapeutics, Janssen, Pfizer, Merck, Ono, and Array, reports receiving commercial research grants from Takeda, Bristol-Myers Squibb, Mirati, Janssen, Pfizer, Novartis, Merck, Ono, and Array, and has ownership interest (including patents) in G1

Therapeutics. K.-K. Wong, F. Li, Q. Huang, and H. Hu have ownership interest in a patent application. No potential conflicts of interest were disclosed by the other authors.

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